

Welcome to

Become a Python Developer

Course Instructor: Md. Azizul Hakim

Lets discuss the problems

- **Leap Year checker**
- **Grades classification**
- **Vowel or consonant**
- **Triangle checker**

loops are used to repeat a block of code

syntax

```
for val in sequence:  
    # statement(s)
```

Example

```
for x in 'Python':  
    print(x)
```

```
#another loop  
for x in range(6):  
    print(x)  
for x in range(2, 6):  
    print(x)  
for x in range(2, 30, 3):  
    print(x)
```

loops are used to repeat a block of code

syntax

while condition:

 # body of while loop

Example

program to display numbers from 1 to 5

initialize the variable

i = 1

n = 5

while loop from i = 1 to 5

while i <= n:

 print(i)

 i = i + 1

Break is used to terminate the loop immediately when it is encountered

continue is used to skip the current iteration of the loop

Example

```
#break
```

```
for i in range(5):
```

```
    if i == 3:
```

```
        break
```

```
    print(i)
```

```
#continue
```

```
for i in range(5):
```

```
    if i == 3:
```

```
        continue
```

```
    print(i)
```

Lets solve this problem using **for** loop

- **Sum of N Numbers:** Write a Python program using a for loop to calculate the sum of the first N natural numbers, where N is taken as input from the user.
- **Table of a Number:** Write a Python program using a for loop to print the multiplication table of a given number N.
-

Lets solve this problem using **while** loop

- **Factorial Calculator:** Write a Python program using a while loop to calculate the factorial of a given number N
- **Count Digits in a Number:** Write a Python program using a while loop to count the number of digits in a given integer N.
-